Customer Shopping Behavior Analysis

Project Report | Python • MySQL • Power BI

Onkar Vijay Gawad | B.Sc. Information Technology, University of Mumbai

1. Introduction

Customer Shopping Behavior Analysis is an end-to-end data analytics project that explores customer purchasing patterns using Python, MySQL, and Power BI. The project prepares shopping data, uses SQL to answer business questions, and presents key measures and breakdowns by product category, age group, and subscription status in an interactive dashboard.

2. Project Objectives

- Prepare customer shopping data by checking data quality and transforming fields.
- Explore purchasing patterns across product categories, customer groups, discounts, and subscriptions.
- Use SQL queries to calculate business metrics and answer analytical questions.
- Present results in a Power BI dashboard with key performance indicators, charts, and filters.

3. Dataset Overview

The dataset contains customer demographics, purchased items, product categories, purchase amounts, locations, sizes, colors, seasons, review ratings, subscription status, shipping type, discount usage, previous purchases, payment methods, and purchase frequency.

Measure	Value
Records	3,900 customers
Columns	18 in the raw data; 19 after preparation
Missing values	37 in Review Rating (about 1%); none elsewhere
Duplicates	0 duplicate rows; 0 duplicate customer IDs
Total revenue	$233,081
Average purchase / rating	$59.76 per purchase / 3.75 out of 5
Date information	None, so the project does not analyze trends over time

4. Tools and Technologies

- **Python and Pandas:** data inspection, cleaning, transformation, and feature engineering.
- **MySQL:** database storage and SQL-based business analysis.
- **Power BI:** interactive dashboard and data visualization.
- **Jupyter Notebook:** Python workflow and analysis.

5. Python Data Preparation

Step	Action	Result
Inspect	Checked structure, data types, missing values, and duplicates	3,900 rows, 18 columns
Missing values	Filled the 37 missing Review Rating values with the mean rating	0 missing values
Column names	Converted to lowercase with underscores; renamed purchase_amount_(usd) to purchase_amount	Consistent names for SQL and Power BI
Age groups	Created age_group using age quartiles	Young Adult 18-31, Adult 32-44, Middle-aged 45-57, Senior 58-70
Purchase frequency	Mapped frequency labels to approximate day intervals (for example Weekly = 7, Monthly = 30)	purchase_frequency_days, all rows mapped
Redundant column	Found discount_applied and promo_code_used identical and removed promo_code_used	19 columns
Export	Saved the cleaned CSV and loaded it into MySQL with SQLAlchemy	3,900 rows in customer_shopping_behavior

6. MySQL Database and Analysis

The prepared dataset was loaded into the MySQL database customer_behavior, in the customer_shopping_behavior table. SQL was used to analyze revenue, average purchase amount, review ratings, discounts, subscriptions, customer segments, and product performance. The queries use aggregation, GROUP BY, subqueries, CASE expressions, and the DENSE_RANK window function.

Business question	Result
Revenue by gender	Male $157,890 (68%); Female $75,191
Discount users spending above the average purchase	839 customers, averaging $79.79 against an overall average of $59.76
Highest average review ratings by product	Gloves 3.86, Sandals 3.84, Boots 3.82 (differences between products are small)
Average purchase by shipping type	2-Day Shipping highest at $60.73; Standard lowest at $58.46
Subscribers vs non-subscribers	1,053 subscribers: $62,645 revenue, $59.49 average. 2,847 non-subscribers: $170,436 revenue, $59.87 average
Products with the highest discount usage	Hat 50.0%, Sneakers 49.7%, Coat 49.1% of orders
Segments by previous purchases	Loyal (more than 10): 3,116; Returning (2 to 10): 701; New (1): 83
Top products within each category	Accessories: Jewelry; Clothing: Blouse; Footwear: Shoes; Outerwear: Coat
Subscription status among repeat buyers (more than 5 previous purchases)	958 subscribers and 2,518 non-subscribers, so subscribers are about 28% of repeat buyers
Revenue contribution by age group	Young Adult 26.7%, Middle-aged 25.4%, Adult 24.0%, Senior 23.9%

7. Power BI Dashboard

An interactive Power BI dashboard summarizes the dataset and lets users explore the results through slicers. It includes:

- **KPI cards:** Total Customers, Total Revenue, Average Purchase Amount, and Average Review Rating.
- **Donut chart:** customer subscription status.
- **Column chart:** revenue by product category.
- **Bar chart:** revenue by age group.
- **Slicers:** subscription status, gender, category, and shipping type.

8. Key Findings

- **Revenue is concentrated in two categories.** Clothing earns $104,264 (about 45%) of the $233,081 total. Clothing and Accessories together bring in about 77%.
- **Order volume drives the difference.** The average purchase is $57 to $60 in every category, so the revenue gaps come from the number of orders rather than price.
- **Subscribers (27% of customers) do not spend more.** They spend $59.49 on average against $59.87 for non-subscribers, with the same average rating of 3.75.
- **Age groups are fairly balanced.** Young Adults (18-31) lead at $62,143 and Seniors (58-70) are lowest at $55,763, a gap of about 11%.
- **Discounts did not raise the average purchase.** 43% of orders used a discount, and their average purchase ($59.28) was slightly below non-discounted orders ($60.13).

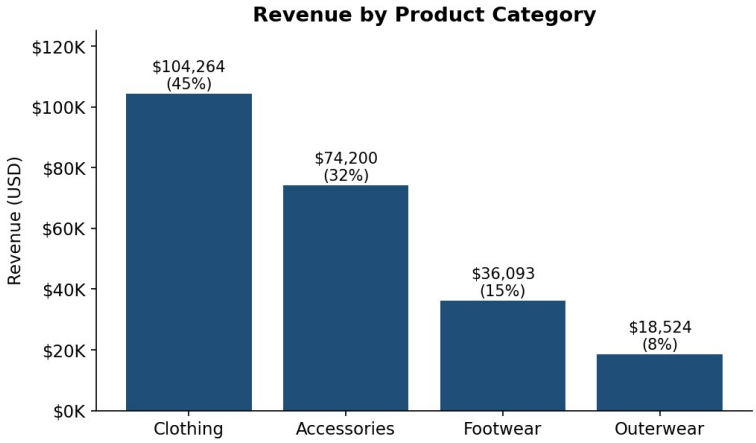


Figure 1: Revenue by product category

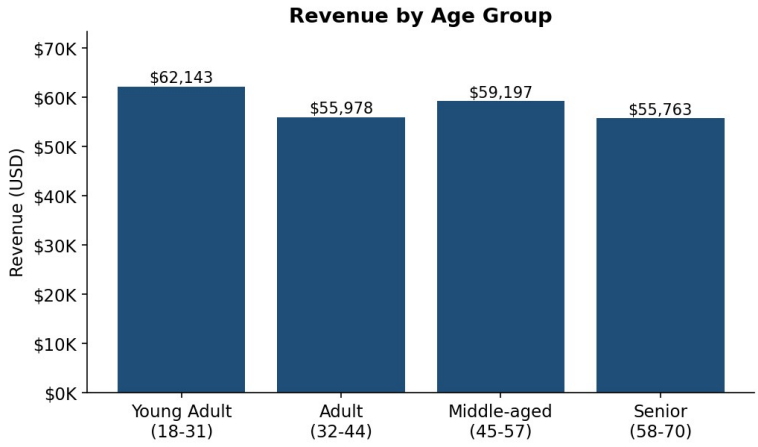


Figure 2: Revenue by age group

9. Data Validation

- Confirmed that the table contains 3,900 records.
- Checked the Review Rating field and confirmed that it contains no NULL values in the loaded table.
- Checked customer_id uniqueness and received zero duplicate IDs.
- Confirmed that revenue by category and revenue by age group each add up to the overall total of $233,081.

10. Limitations and Future Work

- The dataset has no date field, so seasonal totals are available but trends over time cannot be analyzed.
- Missing ratings were filled with the overall mean, which can hide differences between categories.
- The Power BI report uses a single table. Future work includes adding DAX measures (for example revenue share of total) and a data model with separate product and customer tables.
- Future work also includes applying the same workflow to a dataset with order dates to study trends over time.

11. Conclusion

This project demonstrates a practical analytics workflow from data preparation to database querying and dashboard development. Python was used to clean and transform the data, MySQL supported business analysis, and Power BI provided an interactive way to view key customer and purchasing patterns. The main insight is that revenue depends on order volume in Clothing and Accessories rather than on higher prices, and that subscription status shows no link to higher spending in this data.

12. Project Deliverables

- Python notebook for data preparation and loading into MySQL.
- SQL file with the business queries.
- Power BI dashboard file (.pbix) and dashboard screenshots.
- This project report.